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Nasopharyngeal Cancer Treated with Proton Beam Radiotherapy: Patterns of Disease Control and Toxicity with Extended Follow-up

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Purpose/Objective(s): Nasopharyngeal cancers (NPC) are uncommon in the United States and often present with locally advanced disease. While NPC is highly sensitive to chemotherapy and radiation, treatment-related toxicities can be significant given the disease location. Proton beam radiotherapy (PBT) has the potential to reduce dose to normal tissues, however, long-term outcomes and toxicity data on NPC patients treated with PBT are scarce. The purpose of our study is to report outcomes, recurrences, acute and late toxicities, and dose-volume histogram data for a large cohort of NPC patients treated with PBT.

Materials/Methods: Under a prospective institutional review board-approved outcomes study protocol, we reviewed the charts of adult patients (≥ 18 y/o) diagnosed with NPC between 2008 - 2023. Forty-two patients were included (mean age 53), 50% were Epstein-Barr virus positive and 31% were p16 positive. At diagnosis, 50% were T4 and 29% were T3. Fifty-seven percent received PBT only, while 43% received mixed PBT and intensity modulated radiation therapy plans. The primary clinical target volume received a median dose of 7000 cGy/RBE (range, 6120 – 7440). Ninety-five percent of patients received concurrent platinum-based chemotherapy and 62% received induction chemotherapy.

Results: Median follow-up was 4.6 years (range, 0.4-16.2) for all patients, and 5.3 years (range, 0.9-16.2) for living patients. Five-year rates of local control, locoregional control (LRC), distant metastasis-free survival, disease-free survival, cause-specific survival, and overall survival were 92%, 89%, 89%, 80%, 92%, and 75%, respectively. Seven patients had disease progression events, three of which were local recurrences that were all within the primary dose level. Three of the four patients with non-local recurrences were successfully treated with salvage therapy. LRC was worse for T4 patients (76%) compared to T1-3 patients (100%). 55% (n=23) of patients developed a grade 3 toxicity and 2% (n=1) developed a grade 4 toxicity. Thirty-six percent (n=15) developed a late toxicity, the most common of which was hearing loss (n=7). No treatment-related brain necrosis or grade 3+ ocular toxicities occurred.

Conclusion: This study represents one of the largest long-term analyses of locally advanced NPC patients treated with PBT in North America, demonstrating excellent LRC without increased marginal recurrences. While it is possible to spare normal tissue with PBT, serious treatment-related toxicities remain a concern for these patients. These findings support the use of PBT in NPC, but further research is needed to better understand the benefits and cost-effectiveness.

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Larger ¹²⁵I Radiotherapy Plaque Outcomes for Choroidal Melanoma: A Single Institution Retrospective Study

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Purpose/Objective(s): Choroidal melanoma is the most common primary intraocular malignancy in adults. The COMS trial established globe-sparing therapy with ¹²⁵I plaque brachytherapy as a standard of care for small to medium-sized tumors. While plaques in COMS were available up to 20 mm in size, only 16% of tumors were 14 mm or greater. Limited data

exists for larger plaque radiotherapy outcomes. This study aims to report clinical outcomes for patients with larger choroidal melanoma treated with ¹²⁵I brachytherapy. We hypothesize that ¹²⁵I brachytherapy provides effective local control and acceptable toxicity profiles in select patients using large eye plaques.

Materials/Methods: Patients diagnosed with choroidal melanoma between 2005 and 2024 treated with ¹²⁵I brachytherapy using standard COMS 20mm to 22mm plaque sizes were included. Tumor characteristics, ocular anatomy, radiation parameters, and outcomes were extracted from electronic medical record (EMR). Monte Carlo-based dosimetry was used to generate dosimetric parameters. Kaplan-Meier statistics estimated overall survival, local control, and metastases-free survival. Univariate analysis was performed using Cox proportional hazards regression to examine associations between ocular structure doses and adverse events.

Results: 175 patients were included. Baseline characteristics are presented in Table 1. The most common prescription dose was 85 Gy (88%), prescribed to a median depth of 6 mm (IQR: 5 to 7). Median follow up was 8.9 years (IQR: 2.3 to 9.3) and median survival was 9.6 years (CI: 7.2 to 12.8). The 10-year overall survival and metastases free survival was 49% and 62%, respectively. The 10-year local recurrence free survival was 87%. Nineteen (11%) patients underwent enucleation, with 8 (42%) for recurrence and 11 for RT complications including intraocular hemorrhage (3), persistent pain (3), progressive neovascularization (2), retinal detachment (2), or excessive inflammatory response (1). 5-year cumulative incidence of visual acuity decrease of 2 Snellen lines was 77%. Higher doses to ocular structures (ocular disc point and max dose, lens point dose) were associated with higher rates of adverse ocular effects including decreased visual acuity, cystoid macular edema, and radiation maculopathy.

Conclusion: ¹²⁵I plaques of 20 and 22 mm resulted in favorable local control and metastasis-free survival. Higher doses to critical ocular structures were associated with increased toxicity. These findings support the role of ¹²⁵I brachytherapy as an alternative to enucleation for large tumors, with implications for personalized treatment planning to balance tumor control and vision preservation.

Abstract 2718 – Table 1

Patient Demographics	N=174
Age (median)	62 (IQR: 52 to 73)
Male	88 (51%)
Female	86 (49%)
Pre-existing Ocular Conditions	Cataracts: 78 (45%) Glaucoma: 10 (6%)
Largest Basal Diameter of Tumor (mm)	16 (IQR: 15 to 17)
Tumor Thickness (mm)	5 (IQR: 4 to 7)

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Extended-Field Chemoradiation in Cervical Cancer with PET/CT-Detected Para-Aortic Lymph Node Metastases

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Purpose/Objective(s): To evaluate clinical outcomes and identify prognostic factors in patients with para-aortic lymph node (PALN)-metastatic cervical cancer detected by PET/CT who were treated with extended-field radiotherapy and concurrent cisplatin-based chemotherapy.